

Week 2 Competition: Outcome After Treatment

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KAUST Academy & Mawhiba
AI Olympiad Training (Beginners)
Week 2, Day 14, Competition briefing

2 October 2026

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2. How You Are Scored
3. Baselines
4. Where the Points Are
5. Rules and Logistics

Competition: **The Task**

- ▶ **This week:** the line and the sigmoid (Day 08), metrics and the overfitting alarm (Day 09), KNN, SVMs, trees, forests, boosting and tuning (Day 10), the deep dives (Days 11 and 12), and the evaluation routine (Day 13).
- ▶ **Today, 13:00 to tomorrow 13:00:** one real table, one metric, one leaderboard. Teams of up to three. The week's Lab 2 without the scaffolding.
- ▶ **This deck:** the task and its files, how you are scored, the baselines you must beat, where the points are, and the rules.

The one-line brief

Each row is a patient. The hospital wrote down facts when treatment started and, later, how it ended. Predict which test patients end with the worse outcome.

- ▶ When treatment started the hospital recorded: the patient's age and living situation, how big the growth was and how far it had spread, how the cells looked under a microscope, two lab tests, how many tissue samples were checked and how many were affected, and a page of blood measurements.
- ▶ Later it recorded the ending: **outcome 0**, the patient was doing well at the end of the follow-up; **outcome 1**, the worse outcome, the patient did not recover.
- ▶ **You need no medical knowledge.** Every column is explained in plain words on the Kaggle Data tab and in the README; two frames of this deck repeat them.

The task

For each of the 1,127 test patients, write 0 or 1. About one patient in six is a 1.

File	Rows	What it is
<code>train.csv</code>	2,092	one patient per row, 34 feature columns and the outcome column
<code>test.csv</code>	1,127	the same 34 columns, no outcome
<code>sample_submission.csv</code>	1,127	the exact shape of what you upload: id,outcome, all 0
<code>starter.ipynb</code>		loads the data, trains the baseline, writes a valid submission.csv

- ▶ The data is attached to the competition's Kaggle page. In a Kaggle notebook it is already there; on Colab the starter downloads it with `kagglehub` (needs your Kaggle token).
- ▶ `id` is a row number, nothing more. Keep it; it is how your predictions are matched to the patients.

The Columns in Plain Words (1 of 2)

age	years, written with decimals as recorded
population_group	A, B or C: three groups of patients, no order
living_situation	H1 married, H2 single, H3 divorced, H4 widowed, H5 separated
growth_size_class	how big the growth was when found, A1 (smallest) to A4 (largest)
spread_class	how far it reached into nearby tissue, B1 (least) to B3 (most)
size_and_spread	the two classes above written together, for example A2_B1
overall_stage	the doctors' overall stage, C1 (earliest) to C5 (most advanced); blank for some patients
extent	E1 stayed in one area; E2 reached distant parts of the body
cell_appearance	under the microscope: K1 almost normal, K2 somewhat changed, K3 very changed, K4 nothing like normal
cell_grade	the same idea as a number written in text: 1, 2 or 3 (one odd value appears)
growth_size_mm	size in millimetres; blank for some patients, and some values are impossible for a size
test_e, test_p	two lab tests, P positive or N negative; a positive test means a common treatment is expected to work
tests_e_and_p	the two results written together, for example P_N
samples_checked	how many small tissue samples the doctors checked
samples_affected,	how many of those showed the illness, and how many did not
samples_clear	

The Columns in Plain Words (2 of 2): the Blood Page

blood_pressure_high	blood_pressure_low	cholesterol
body_temperature (degrees C)	blood_oxygen (percent)	breaths_per_minute
blood_sugar	bmi (weight for height)	pulse (beats per minute)
creatinine	uric_acid	hemoglobin
kidney_filter_rate	sodium	potassium
albumin	lactate	

- ▶ Seventeen numbers the hospital measured in the blood or at the bedside. You do not need to know what any of them means medically: the only question is whether it helps the model, and Day 13 gave you the tool that answers it.

What usually goes together

A bigger growth, more affected samples, a higher overall stage or more spread go with outcome 1 more often. A positive lab test goes with outcome 0 more often. That is all the medicine you need.

Competition: **How You Are Scored**

Count over the test patients: TP, the ones you marked 1 who were 1; FP, marked 1 but were 0; FN, marked 0 but were 1.

$$\text{precision} = \frac{TP}{TP + FP} \quad \text{recall} = \frac{TP}{TP + FN} \quad F1 = \frac{2 \cdot \text{precision} \cdot \text{recall}}{\text{precision} + \text{recall}}$$

Worked example

100 test patients, 15 of them a real 1. You mark 20 as 1, and 9 of those are right:
 $TP = 9$, $FP = 11$, $FN = 6$.

$\text{precision} = 9/20 = 0.45$, $\text{recall} = 9/15 = 0.60$, $F1 = 2 \cdot 0.45 \cdot 0.60 / 1.05 = 0.514$.

- ▶ The score is computed on your 0/1 labels, not on probabilities. The cut you choose is part of the model (Day 09).
- ▶ Only the 1s count. Being right about the 0s earns nothing by itself; missing a 1 or inventing one costs.

Everyone 0: $TP = 0$, so precision and recall are 0 and $F1 = \mathbf{0.000}$. The sample submission does exactly this.

Everyone 1: every real 1 is found (recall = 1), but precision is only the share of 1s, $p = 0.154$:

$$F1 = \frac{2 \cdot p \cdot 1}{p + 1} = \frac{0.308}{1.154} = \mathbf{0.267}$$

The floor is 0.267, not 0

A model that scores below 0.267 is worse than writing 1 everywhere. The first thing to check about any submission is that it beats this number on your own folds.

The baseline

The starter's logistic regression scores about 0.385 on 5-fold cross-validation and 0.396 on the public board.

- ▶ **Public leaderboard:** 40% of the test patients, scored on every upload, visible all day.
- ▶ **Private leaderboard:** the other 60%, revealed at 13:00 tomorrow. It is the final ranking.
- ▶ You choose **2 final submissions** on the Kaggle page before the end; if you do not, Kaggle takes your best public scores. Choose them by your cross-validation, not by the board (Day 13).

The second decimal is partly luck

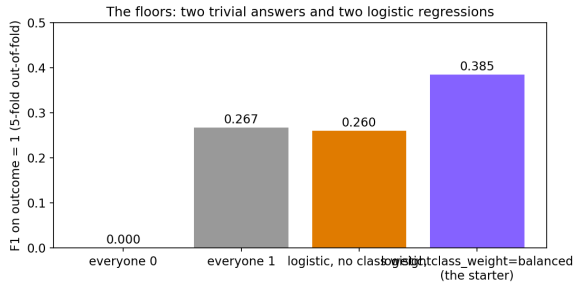
The test set holds about 170 patients with outcome 1. F1 on that many moves by about 0.03 between samples. A public score 0.02 above a rival's says little; a cross-validated gap seven spreads wide says a lot.

Competition: **Baselines**

1. **Data loading (do not modify):** finds the data on Kaggle, on Colab or on a laptop and prints the shapes and the outcome rate.
2. **Look at the table:** head, dtypes, blanks, describe, value_counts of three text columns. Read every line; the table has surprises.
3. **Baseline:** median imputer and scaler on the numbers, most-frequent imputer and one-hot on the text, logistic regression with `class_weight="balanced"`, 5-fold stratified F1 printed, then fit on all rows.
4. **Submission generator (do not modify):** checks the length and that every value is 0 or 1, writes id,outcome.

```
scores = cross_val_score(model, X, y, cv=StratifiedKFold(5, shuffle=True,
    random_state=42), scoring="f1")
# 5-fold F1: [0.379 0.406 0.393 0.379 0.366]    mean 0.385    spread 0.014
submission = make_submission(test_pred)      # wrote submission.csv: 1127
rows, share of 1 = 0.302
```

Run it top to bottom, upload the file: on the board within fifteen minutes (Day 06's rule).



Source: the competition's `train.csv`, 5-fold out-of-fold predictions; `scripts/day14_comp_figures.py`.

- ▶ Everyone 0: 0.000. Everyone 1: 0.267.
- ▶ Logistic regression on the raw columns, no class weight: **0.260**, below “everyone 1”. It predicts 0 for almost every patient, because 0 is right 85% of the time.
- ▶ The same model with `class_weight="balanced"`: **0.385**. One argument, twelve points.

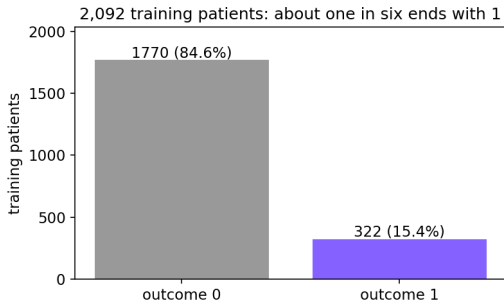
- ▶ The loss the model minimises treats every patient the same. With 85% zeros, the cheapest way to be mostly right is to say 0 to everyone; the rare 1s are not worth the trouble.
- ▶ `class_weight="balanced"` multiplies each 1 by $0.5/0.154 = 3.2$ and each 0 by $0.5/0.846 = 0.6$ in the loss (Day 11). The model now has to find the 1s.
- ▶ What it changes: recall of the 1s goes up, precision goes down, F1 goes up. What it does not change: the ranking of the patients (ROC-AUC stays the same). It is a threshold move built into `fit`.

Takeaway

On this task, how you treat the rare class decides more than which model family you pick. Every model you try needs its own answer to the imbalance: a class weight, a threshold, or both.

Competition: **Where the Points Are**

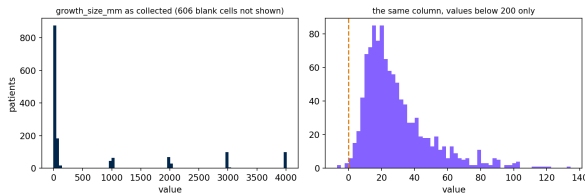
Problem 1: the Outcome Is Rare



Source: the competition's `train.csv`; `scripts/day14_comp_figures.py`.

- ▶ 322 ones among 2,092 patients. Every fold must keep that share:
`StratifiedKFold(5, shuffle=True, random_state=...)`, and `stratify=y` in any hold-out (Day 13).
- ▶ Report the spread of your folds next to the mean. With 64 ones per fold, F1 moves by a few hundredths on its own.
- ▶ Accuracy is meaningless here (0.85 for doing nothing). Score everything with `scoring="f1"`.

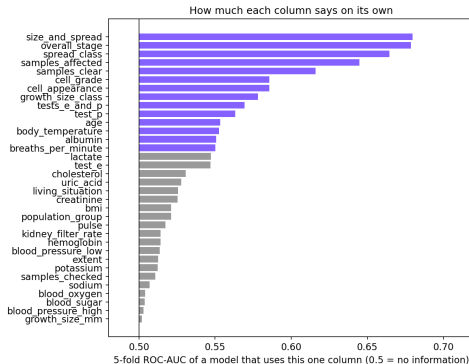
Problem 2: the Table Is As Collected



Source: the competition's `train.csv`, column `growth_size_mm`; `scripts/day14.comp.figures.py`.

- ▶ 606 blank sizes, 410 values above 900 in three narrow spikes, 5 below zero. Day 06's rule: a value that cannot be a measurement is a *code*, not a number. What does each spike mean, and what should the model see instead?
- ▶ Sample counts go negative too, one text column has a value with a space at the end, and `cell_grade` holds a word where a number should be; `value_counts(dropna=False)` on every text column finds all of it in two minutes.
- ▶ Decide once, in one function, applied to train and test alike (Day 12).

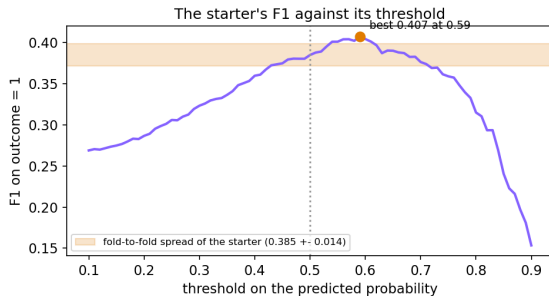
Problem 3: Not Every Column Speaks



Source: the competition's `train.csv`, one small model per column, 5-fold ROC-AUC; `scripts/day14_comp_figures.py`.

- ▶ The stage, spread and sample columns say the most on their own (0.62 to 0.68). Most of the blood page sits at 0.50 to 0.55, which is what a column of random numbers scores.
- ▶ A column that says nothing on its own can still help in combination; a column that is noise can only hurt a line and distract a tree. Day 13's permutation importance on a validation split separates the two.
- ▶ Fewer, cleaner columns also make every cross-validation run faster: more experiments in 24 hours.

Problem 4: the Threshold Is Yours to Choose



Source: the starter's out-of-fold probabilities on `train.csv`; `scripts/day14-comp-figures.py`.

- ▶ predict cuts at 0.5. The curve shows the starter's F1 for every cut: a peak of 0.407 at 0.59 against 0.385 at 0.5.
- ▶ The orange band is the fold-to-fold spread. The peak is a best-of-81 search on 322 ones; part of it is luck. Day 13's rule: a gain smaller than the spread is not a gain until it repeats on other folds.
- ▶ Choose the cut on out-of-fold predictions, never on the training fit, and check it with a second `random_state`.

Model	Needs	What to watch
logistic regression (the starter)	scaling, weight	class C, the cleaned columns, the threshold
random forest	nothing	<code>class_weight="balanced_subsample"</code> , <code>min_samples_leaf</code> , the OOB score for free
gradient boosting	nothing	its probabilities are not balanced: the threshold matters more than for the others
KNN, SVM	scaling	slow to improve here; try only after the three above
a blend of two	out-of-fold predictions	average probabilities, then cut at the OOF-chosen threshold

- ▶ One `StratifiedKFold` object, one scoring function, every model through both. A model with a different fold draw is not comparable.
- ▶ Save the out-of-fold and test predictions of every model you run; blends and threshold choices then cost nothing (Day 13's notebook shape).

The Day 13 routine

1. Run the starter, upload, you are on the board.
2. Look at the table; clean in one function.
3. Stratified folds; out-of-fold predictions per model.
4. Compare models against the spread; prune columns.
5. Threshold and blend on OOF only.
6. Refit the winner on all 2,092 rows; predict test.
7. Upload; pick the 2 final submissions by CV.

Before you click Submit

- ▶ 1,127 rows, columns id, outcome, values 0 or 1 only, no blank.
- ▶ The share of 1s is near 0.15 to 0.35; a file with 0.02 or 0.90 is a bug.
- ▶ The same cleaning ran on test as on train; the same columns in the same order.
- ▶ Your CV number is written down next to the upload, so tomorrow's private score has something to be compared with.

Competition: **Rules and Logistics**

What	Riyadh time	UTC
competition opens	Fri 2 Oct, 13:00	10:00
submissions-per-day counter resets	Sat 3 Oct, 03:00	00:00
last moment to form or merge a team	Sat 3 Oct, 07:00	04:00
competition closes, private board revealed	Sat 3 Oct, 13:00	10:00
notebook hand-in closes (Day 14 form)	Sat 3 Oct, 13:30	10:30

- ▶ Teams of at most 3; solo is allowed; one team per student. Merge on the Kaggle page, not by sharing files.
- ▶ 12 submissions per team per Kaggle day, so 24 in the window; 2 final submissions.
- ▶ Allowed: pandas, NumPy, scikit-learn, matplotlib and everything taught this week. No other data, no searching for where the table comes from.

- ▶ **Your own code.** Discuss ideas with other teams if you like; every line in your notebook is written by your team.
- ▶ **The notebook counts.** One notebook per team through the Day 14 Google Form by 13:30 Saturday, team name in the first cell. A leaderboard score without a notebook is not ranked.
- ▶ **Final ranking:** the private leaderboard. Ties: the earlier of the two final submissions wins.
- ▶ **Do not edit** the two cells marked “do not modify”; they keep your file in the shape the scorer expects.

Links

Join: <https://www.kaggle.com/t/0bc9abf7444a41588f33f496cfcd2bd5>

Brief, column table, starter:

github.com/KAUST-Academy/Olympiad2027/tree/main/L1/Comps/Week2_ML_Techniques

Hand-in form: shared in the course channel.

Good luck

Beat 0.267, then beat 0.385, then beat each other. Write your CV number down every time.